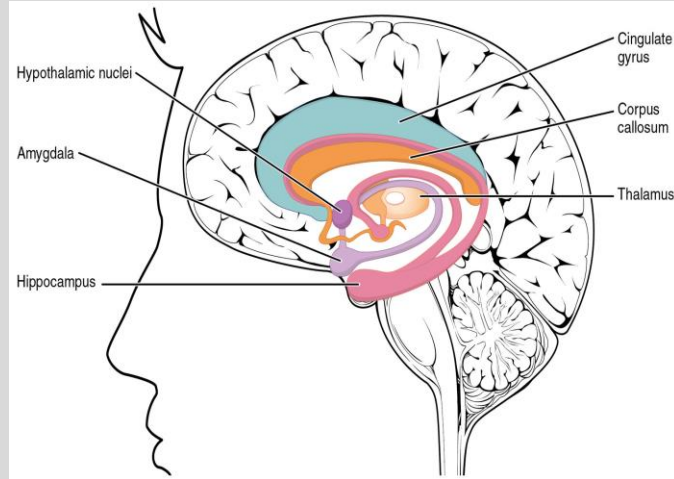


# Session Objectives

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1. Basic anatomy and cellular circuitry of the limbic system
2. Cingulate gyrus
3. Hippocampal formation and memory
4. Amygdala and emotion
5. Different kinds of memory disorders
6. Clinical syndromes associated with limbic system pathology



See **detailed objectives** in CPG  
**PLEASE READ THE Scholar Rx Brick!**

# Session Objectives Mapped to COMLEX Blueprint

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- **Dimension 2**
  - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
    - Amnesia
    - Wernicke-Korsakoff encephalopathy
    - Learning disorders
    - Trauma and stressor-related disorders (eg, adjustment disorders, post-traumatic stress disorder)

# Session Objectives Mapped to USMLE Blueprint

## V. Nervous System & Special Senses

### A. Normal Processes

- 2. Organ structure and function
  - iii. Brain
    - c. limbic system and emotional behavior circadian rhythm sleep-wake disorder

### B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

- 2. Anxiety disorders
  - i – ix: acute stress disorder; anxiety disorders (generalized anxiety disorder, anxiety due to another medical condition, social phobia); hyperventilation syndrome; obsessive-compulsive disorder; panic disorder with and without agoraphobia; phobic disorders; post-traumatic stress disorder; separation anxiety disorder; substance-induced anxiety disorder, trichotillomania
- 3. Mood disorders
  - i – xi: major depressive disorder with and without psychotic features, with and without seasonal pattern; major depressive disorder, postpartum, with and without psychotic features, including screening; cyclothymic disorder; persistent depressive disorder (dysthymia); bipolar disorder, manic/depressed/mixed; premenstrual dysphoric disorder; bipolar and related disorder or depressive disorder due to another medical condition; substance/medication-induced bipolar and related disorder or depressive disorder (illegal or prescribed); suicidal ideation/attempt

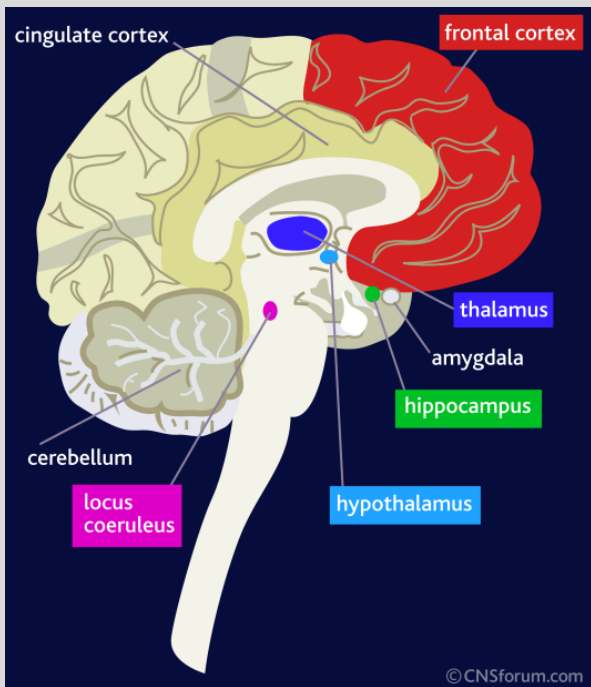
# The Limbic System

Consists of a core of cortical and diencephalic structures to provide primitive **emotional** regulation of the **hypothalamic** and **brainstem autonomic** output.

Latin “Limbus” = “Border”:

- Bordering the cerebral cortex and diencephalon

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## Composition under debate

Central components unanimously agreed on (the “**big 3**”):

- **Cingulate gyrus**
- **Hippocampal formation**
- **Amygdala**

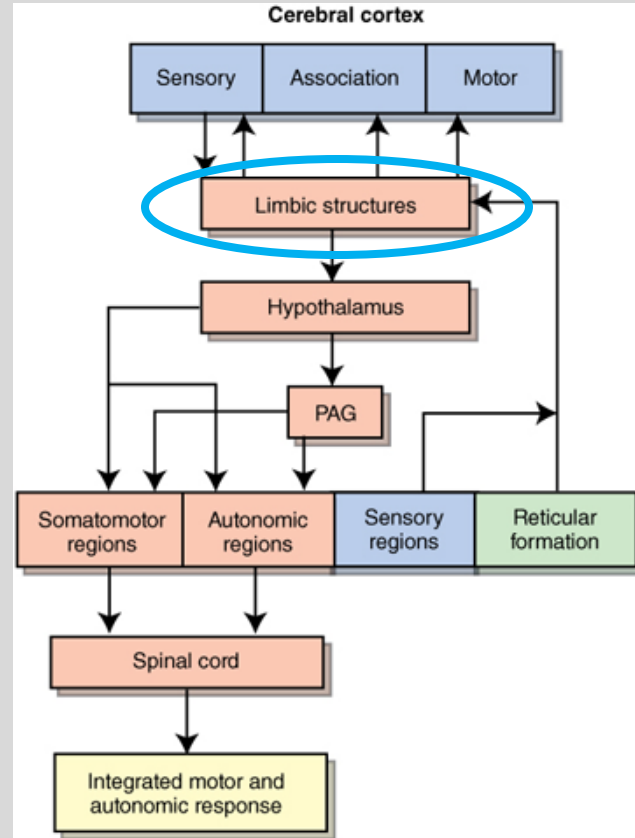
Other areas:

- **Entorhinal cortex**
- **Hypothalamus**
- **Mammillary body**
- **Anterior thalamic nuclei**
- **Dorsomedial nucleus of the thalamus**
- **Orbitofrontal cortex**
- **Septal area**
- **Prefrontal cortex**
- **Insula**

# Connections of Limbic System

Receives **sensory information** from the cortex, spinal cord, and brainstem reticular formation.

Integrate the sensory information with emotional responses and send the primary output to **hypothalamus** to produce autonomic changes.



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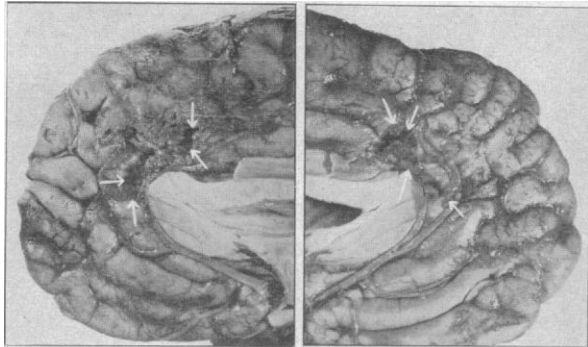
# Cingulectomy for severe anxiety and depression

*Section of Psychiatry with Section of Neurology*

## **Effects of Anterior Cingulectomy in Man<sup>1</sup>**

By C. W. M. WHITTY, D.M., M.R.C.P.

*Department of Neurology, Radcliffe Infirmary, Oxford*



Improved psychiatric symptoms, but with reduced frontal lobe inhibition

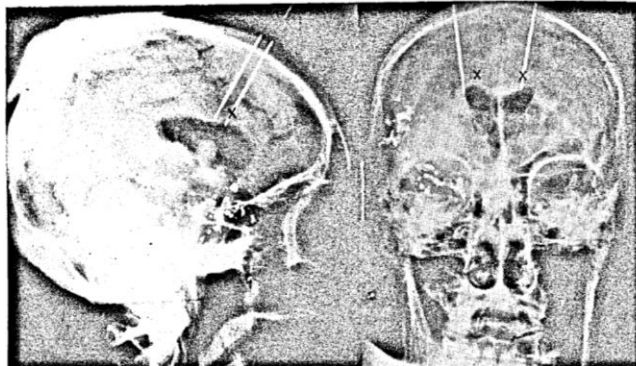
# Cingulumotomy for Intractable Pain

PAIN “RELIEF” BY FRONTAL CINGULUMOTOMY\*

ELDON L. FOLTZ, M.D., AND LOWELL E. WHITE, JR., M.D.

*Division of Neurosurgery, University of Washington, Seattle, Washington*

(Received for publication May 19, 1961)



“I can still feel pain”, but “it was not distressing”, “doesn’t bother me any more”

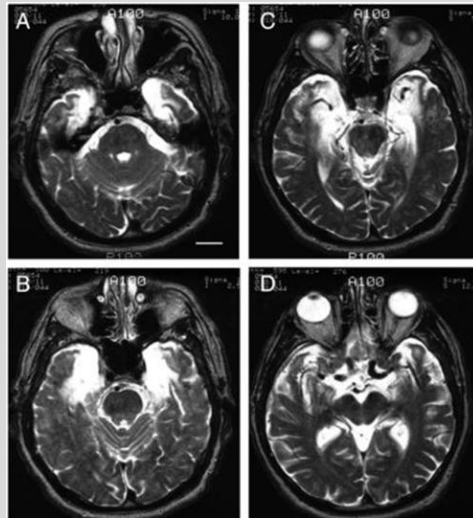
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# Herpes Encephalitis

This patient became profoundly **amnesic** in 1992 after viral encephalitis, which damaged his medial temporal lobe bilaterally.

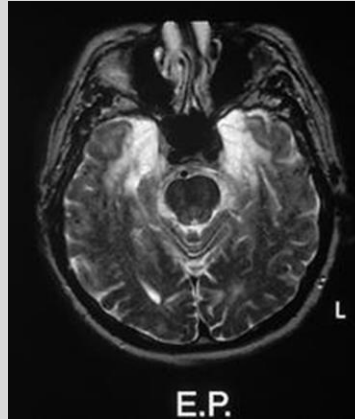
The **herpes simplex virus** has a tropism for the **limbic cortex** and can cause severe encephalitis involving primarily limbic cortical areas



Images are arranged from ventral (A) to dorsal (D), indicating the extent of damage in E. P.'s anterior temporal lobes (bright signal)

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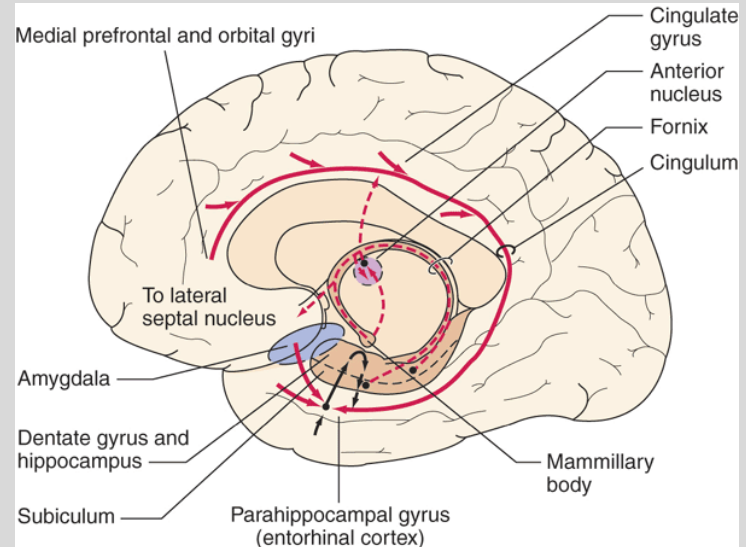
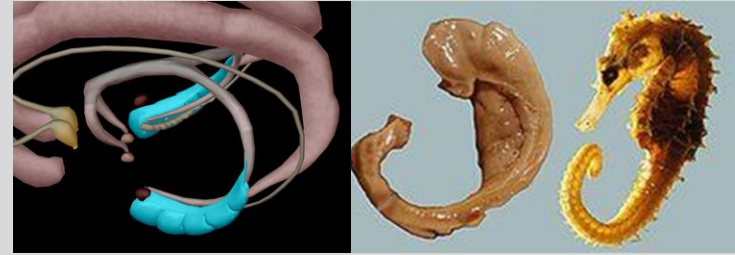
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# Hippocampal Formation

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- Functions in **learning**, **memory** and **recognition of novel objects** and **navigation**.
- It receives major **input** through the **entorhinal cortex** and projects major **output** through the **fornix** to reach **diencephalon**.
- Composed of three major structures:
  - Dentate gyrus
  - Hippocampus
  - Subiculum





# Temporal Lobe Epilepsy

## *Temporal Lobe Epilepsy*

- The temporal lobe (especially the hippocampus and amygdala) has a **lower threshold for epileptic seizure** activity than the other cortical areas
- Seizures that originate in these regions are called **complex partial (psychomotor) seizures**
- Temporal lobe epilepsy may include **abnormal sensations, repeated involuntary movements** such as chewing, swallowing, and lip smacking, disorders of consciousness, **memory loss** and **hallucinations**

# Amygdala

Almond shaped mass of nuclei located deep in the temporal lobes

It is a limbic structure that is involved in many of our **emotions** and **motivations**, particularly those that are related to **survival**

*It is important in processing of **emotions** such as **fear, anger and pleasure***

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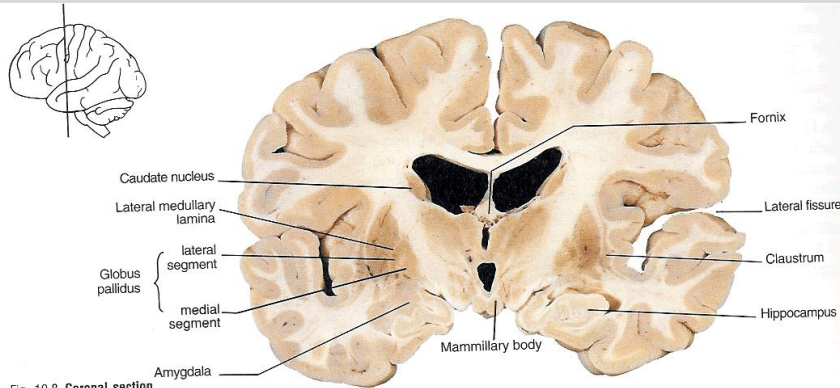
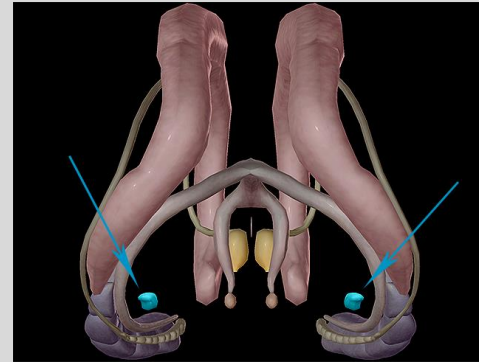


Fig. 10.8 Coronal section.



# Amygdala and Fear

## *Kluver-Bucy Syndrome*

**Bilateral** lesions of the **amygdala**

These lesions result in:

**Placidity**: decrease in aggressive behavior

**Psychic blindness**: objects in visual field are treated inappropriately. Example: Monkeys approach snake with inappropriate docility

**Hypermetamorphosis**: visual stimuli are repeatedly approached as though they were completely new

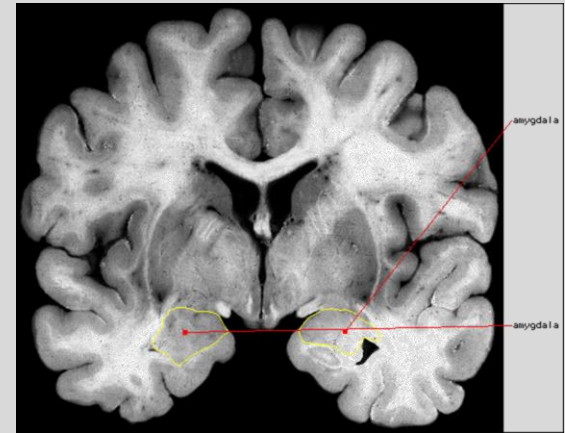
**Hyper-orality**: Increased oral exploratory behavior

**Hypersexuality** and loss of sexual preference

**Anterograde amnesia**: deficit in forming new memories

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# Post-traumatic Stress Disorder (PTSD)

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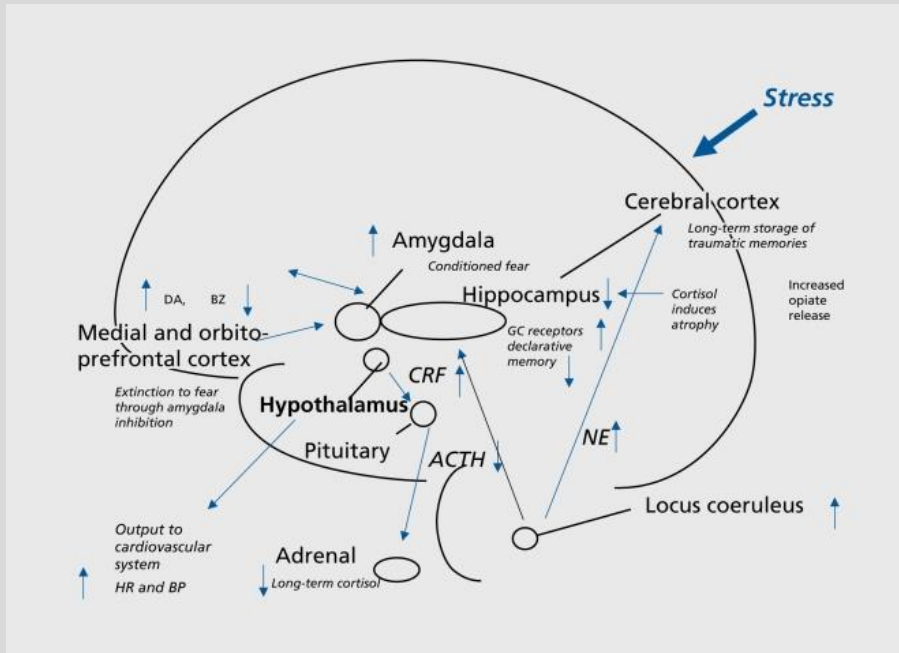


Figure 1. Lasting effects of trauma on the brain, showing long-term dysregulation of norepinephrine and Cortisol systems, and vulnerable areas of hippocampus, amygdala, and medial prefrontal cortex that are affected by trauma. GC, glucocorticoid; CRF, corticotropin-releasing factor; ACTH, adrenocorticotropin hormone; NE, norepinephrine; HR, heart rate; BP, blood pressure; DA, dopamine; BZ, benzodiazapine; GC, glucocorticoid

- PTSD is often characterized by **aberrant amygdala activation** and functional abnormalities in **cortico-limbic circuitry**
- Abnormalities in amygdala activity may be associated with **hyper-arousal** and **hypervigilance** to threat

# Septal Area

It has reciprocal connections with the **hippocampal** formation through the fornix; and with the **hypothalamus** through the medial forebrain bundle

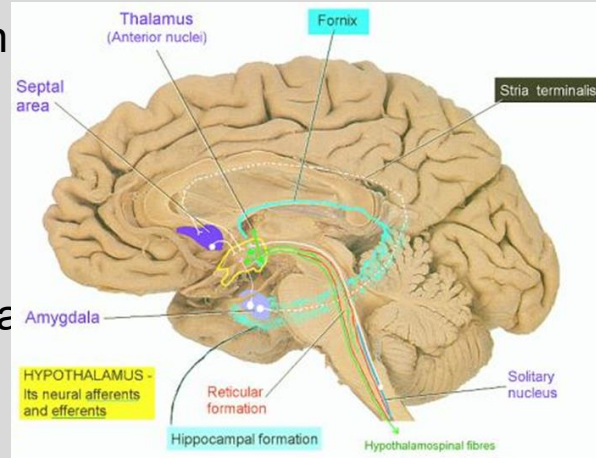
It serves as a **relay** for transmission of hippocampal impulses to the hypothalamus and as a feedback system to the hippocampal formation

In addition, activity in the septal area appears to be important in **reward** and **learning** (**dopamine mesolimbic** pathway)

- **Nucleus accumbens**
- Bed nucleus of the stria terminalis

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# Question

The Syndrome of Wernicke-Korsakoff is characterized by ataxia/nystagmus, ophthalmoplegia, and amnesia, confabulation, etc., respectively due to a lack of thiamine (Vitamin B1) from excessive alcohol consumption and malnutrition. The part of the CNS that is thought to be damaged/decreased activity is:

- A) the hippocampus
- B) the frontal cortex
- C) the parahippocampal gyrus
- D) the mammillary bodies**
- E) the hypothalamus



# Wernicke-Korsakoff Syndrome

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**Vitamin B1 deficiency** in  
alcoholics

## Wernicke Encephalopathy

- Confusion
- Ataxia
- Nystagmus
- Ophthalmoplegia

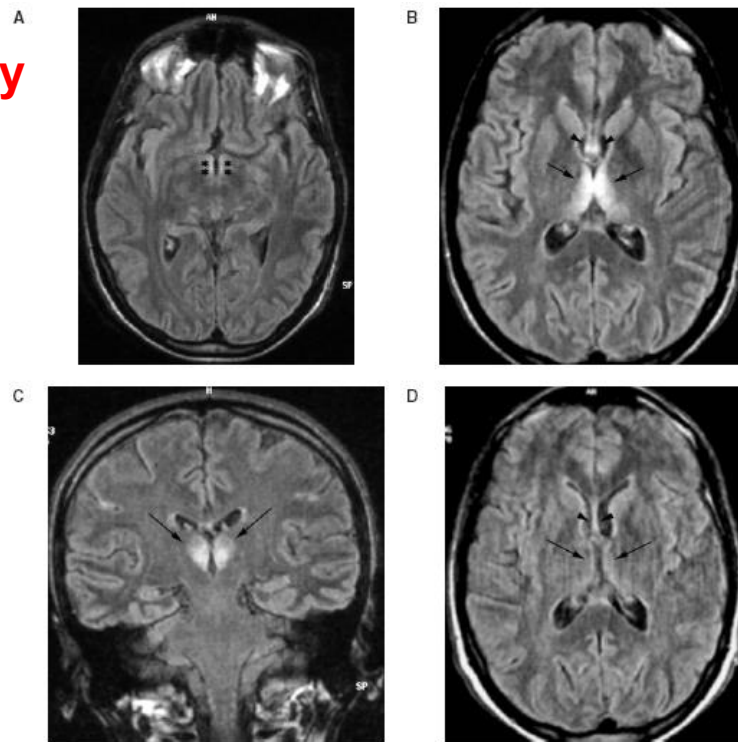
“CAN Of beer” problem

## Korsakoff Psychosis

- Amnesia (anterograde and retrograde)
- Hallucinations
- Confabulations

Treatment:

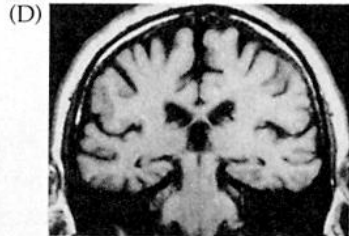
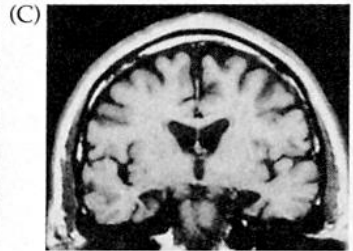
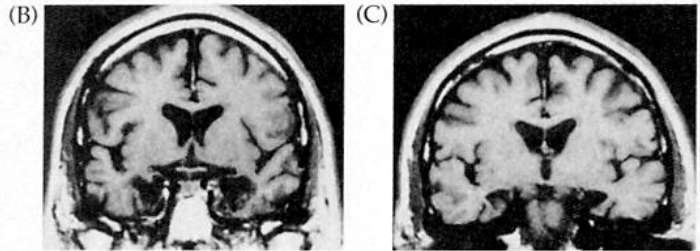
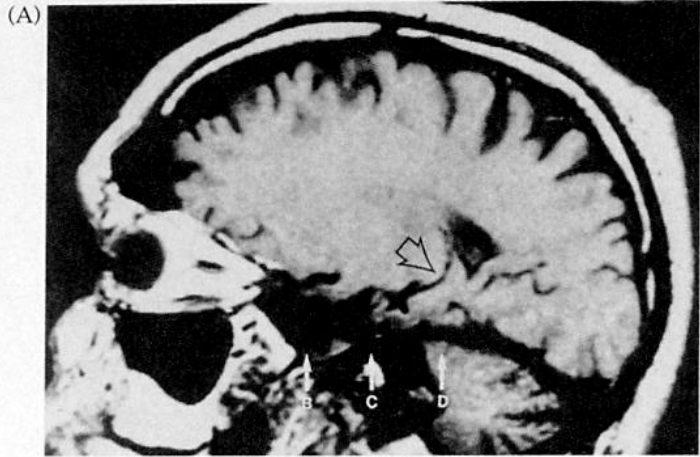
- Thiamine (Vitamin B1) 600 mg/day
- Reversible with early intervention
- Irreversible later on



**Fig. 2** Images from axial and coronal FLAIR sequence of the WK subject. Seven-day MRI study demonstrates signal abnormalities of the mammillary bodies (A) (stars), the medial aspect of the thalami (B, C) (arrows) and the fornix (B) (arrowheads). Five months later signal changes have almost completely disappeared (D) (arrows and arrowheads).

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MRI of patient H.M. - much of the medial temporal lobes removed for treatment of intractable seizures.

H.M. had an inability to transfer short term memory to long term memory for events and facts.

([anterograde amnesia](#))

He could remember events and things that were consolidated years prior to the surgery.

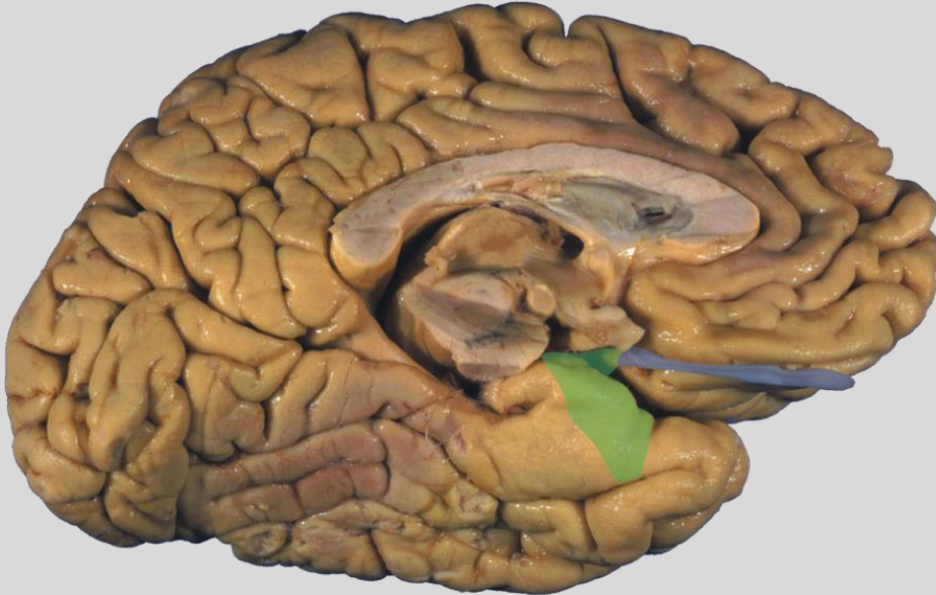
**What else did he lack?**

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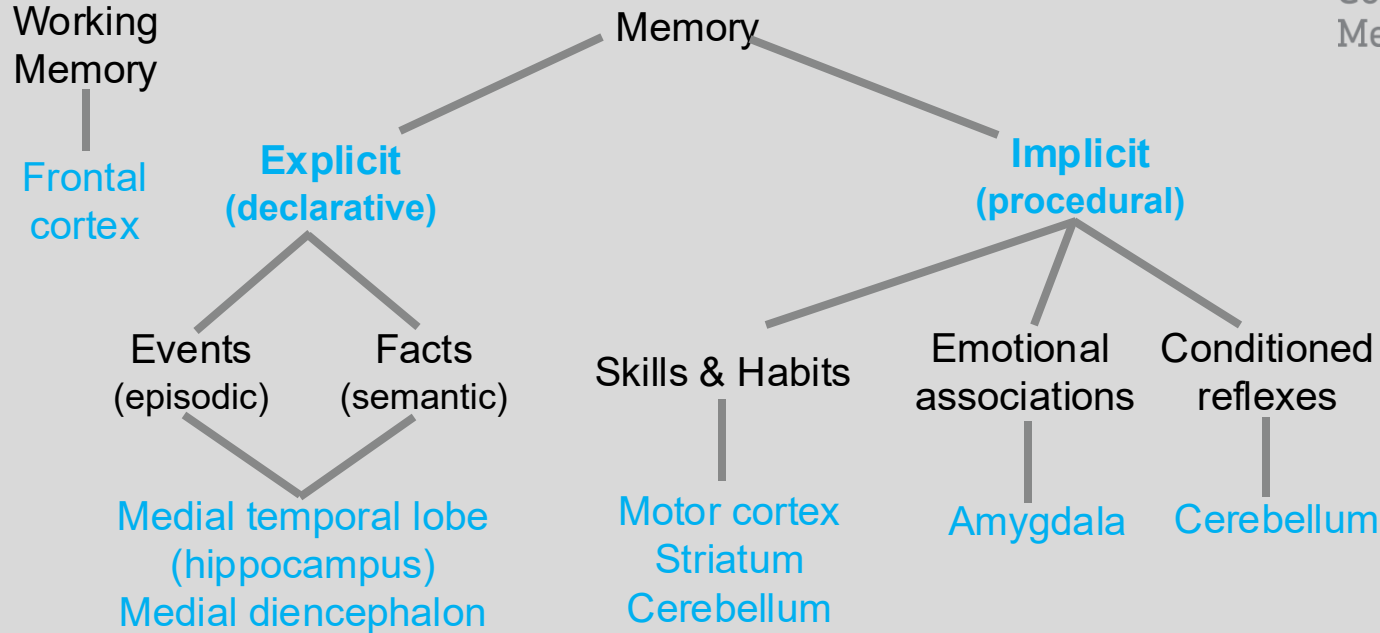
- He doesn't have normal **olfactory** threshold.....he is **unable** to **discriminate** among different **odors** or identify anything by smell – couldn't remember smells.
  - **Piriform cortex** (including the **entorhinal cortex** and **amygdala** mediates the perception of different odors and related emotional responses.
- He is impaired in **keeping track of his internal states**, presumably because of the removal of both **amygdala**.



# Different parts of the brain are responsible for different types of memory

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Long-term memory storage is thought to be in  
diffuse areas of the Association Cortices

# Some Clinical Causes of Amnesia

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| Causes                                                          | Site of Damage                                                            |
|-----------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Vascular</b><br>( <b>PCAs</b> )                              | <b>Bilateral Medial Temporal Lobe</b> (more sensitive to <b>hypoxia</b> ) |
| <b>Trauma</b><br>(concussion, cerebral contusion)               | <b>Mammillary bodies, Mammillothalamic tract, Medial Thalamus</b>         |
| <b>Surgery</b><br>(for temporal lobe epilepsy)                  | <b>Bilateral Medial Temporal Lobe</b>                                     |
| <b>Infections</b><br>(herpes simplex virus)                     | <b>Bilateral Medial Temporal Lobe</b>                                     |
| <b>Vitamin B1 Deficiency</b><br>(Wernicke-Korsakoff's syndrome) | <b>Mammillary bodies, Medial Thalamus</b>                                 |

In all cases, the patients have bilateral injury to some component of their medial temporal lobe or thalamus.

# Summary

| Limbic Structure                | Major Functions                   |
|---------------------------------|-----------------------------------|
| Cingulate cortex                | Depression, anxiety, pain         |
| Amygdala                        | Fear, olfaction                   |
| Hippocampus                     | Memory, learning                  |
| Septal area (Nucleus accumbens) | Reward                            |
| Orbital frontal cortex          | Secondary olfactory cortex        |
| Hypothalamus                    | Homeostasis, autonomic regulation |